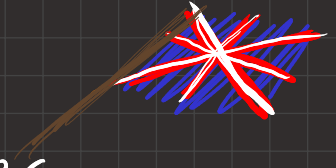
BIOMETRIC SYSTEMS



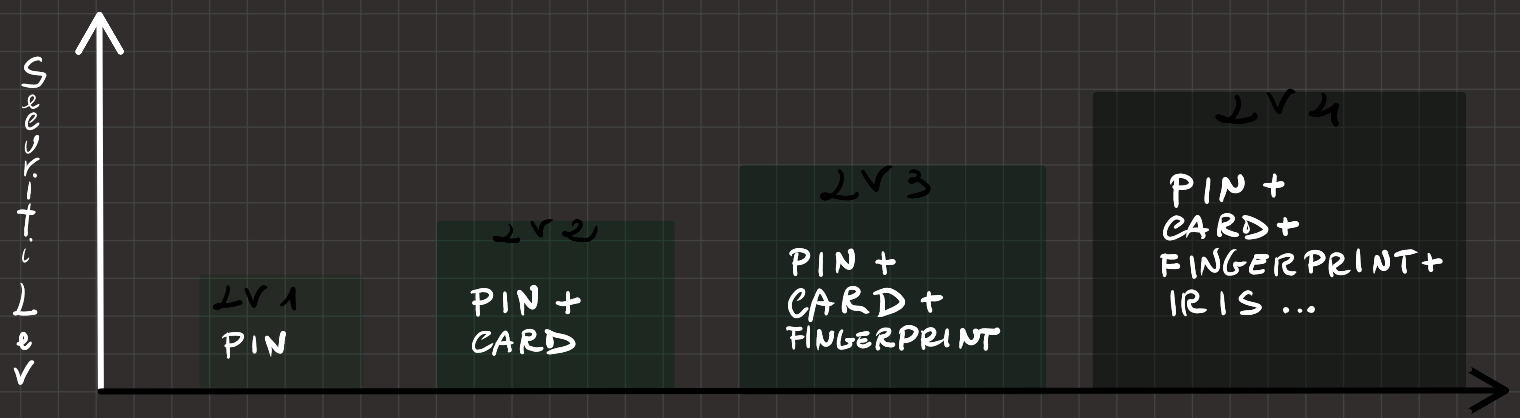
Carlo Re Rome NOV. 10. 2026

FUNDAMENTAL CONCEPTS

- THE TERM BIOMETRICS DERIVES FROM BIOS (LIFE) AND METRON (MEASURE IN GREEK).
- BIOMETRICS REFERS TO STUDY AND USE METHODS FOR DETECT AND MEASURE BIOLOGICAL CARACTERISTICS.
- A BIOMETRIC SYSTEM IS A SPECIALIZED FORM OF PATTERN RECOGNIZING.

WHY BIOMETRIC SYSTEMS?

- BIOMETRIC SYSTEMS ARE USEFULL TO ACCESS SOMETHING FOR EXAMPLE BECAUSE OTHER METHODS HAVE PROBLEMS... A PASSWORD OR A PIN CAN BE STOLEN OR FORGOTTEN, A CARD OR A PASSKEY... THE SAME, BIOMETRIC KEY IS USEFULL BECAUSE YOU ARE THE BIOMETRIC KEY, YOU CAN'T LOST IT OR FORGOT IT BUT BIOMETRIC SYSTEMS ARE NOT 100% SURE BECAUSE THEY CAN MAKE FAKE POSITIVE. A GOOD IDEA IS TO MIX THE THECNOLOGIES.



- A BIOMETRIC SYSTEM IS A SPECIALIZED FORM OF PATTERN RECOGNIZING.

WHY?

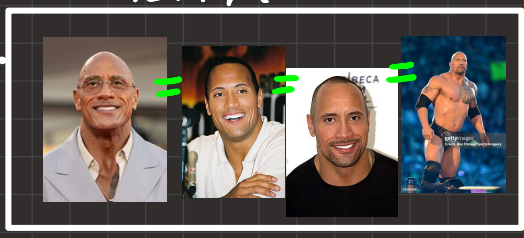
BECAUSE TO RECOGNIZE A FACE, A IRIS ...
EACH AQUISTION IS TRADUCED IN A FEATURE VECTOR, A VECTOR THAT DESCRIBE THE PHOTO...

NOW THE QUESTION IS:
WHAT FEATURE IS RELEVANT?
FEATURES THAT NOT CHANGE A LOT CHANGING THE FOTO OF THE SAME THING..

TWO OBJECTS ARE CONSIDERED THE SAME IF THEIR FEATURE VECTORS ARE SIMILAR AND NOW THE QUESTION IS ANOTHER
WHAT IS THE MARGIN?
THE ANSWER IS: A COMPROMISE.

NOW WE HAVE TO DISTINGUISH 2 VARIABILITY.

- INTRACLASS VARIABILITY .....→
  SAME OBJECT, DIFFERENT SCENARIO



- INTERCLASS VARIABILITY .....→
  DIFFERENT OBJECTS

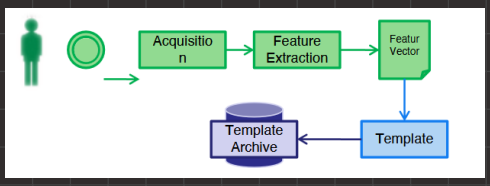


SOFTWERE ARCHITECTURE OF A BIOMETRIC SYSTEM.

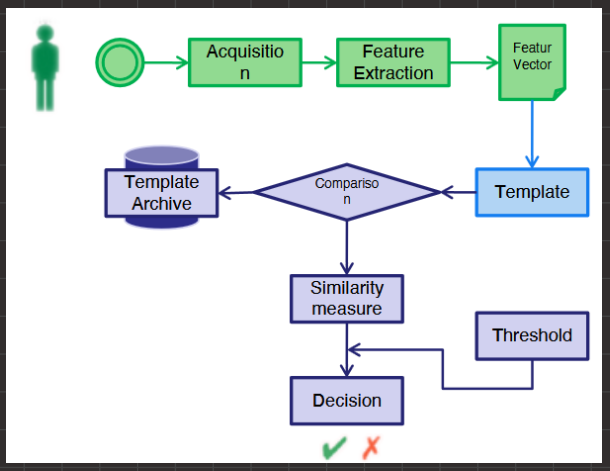
- ENROLLMENT
  THE PART THAT AQUIRE IMAGES OR OTHER FROM A SENSOR, IT TRASFORM IT TO A VECTOR AND TRASFORM THE VECTOR IN A TEMPLATE, IF IT KNOW THE LABEL, SAVE THE TEMPLATE
- RECOGNITION
  THE PART THAT AQUIRE IMAGES OR OTHER FROM A SENSOR, IT TRASFORM IT TO A VECTOR

AND TRASFORM THE VECTOR IN A TEMPLATE, IT DOES A COMPARISON WITH OTHER TEMPLATES AND GIVE US A LABEL (IN SIMPLE WORDS) OR MAKE A DECISION.

- ENROLLMENT



- RECOGNITION



- PROBE, TEMPLATE AND GALLERY:

Probe: each template which is submitted for recognition.

Gallery: the set of templates pertaining to enrolled subjects

TEMPLATE: FEATURE VECTOR OR A COMPRESSED VERSION OF IT

- SCORE AND TRESHOLD.

IF WE COMPARE A PROBE WITH A TEMPLATE IN THE GALLERY WE RECEIVE A SCORE, THE MATHING SCORE.

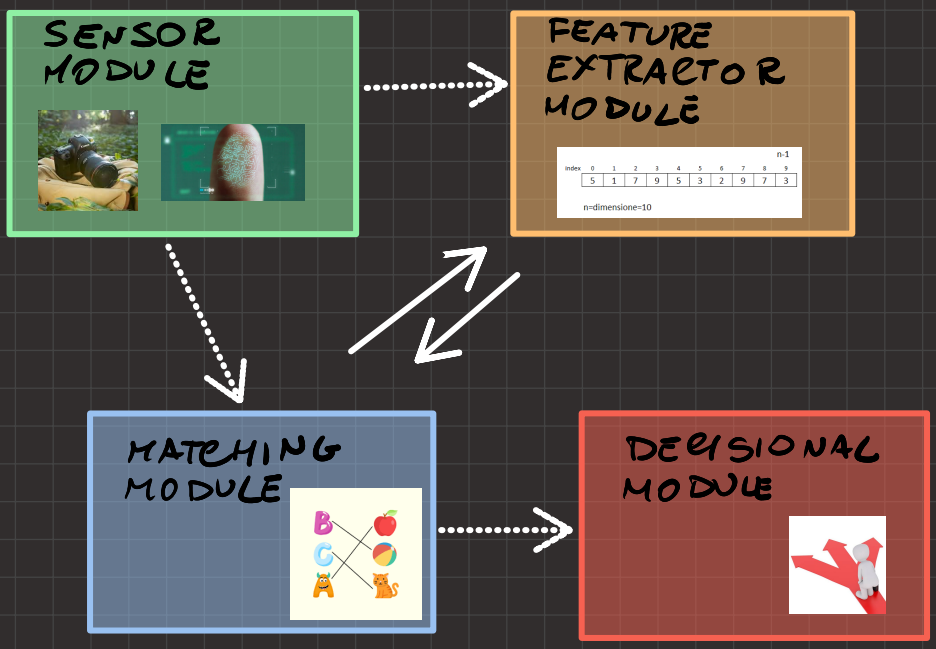
- SIMILARITY $s(p, g_i)$
- DISTANCE $d(p, g_i)$

THE MORE HIGH IS THE SIMILARITY, THE MORE TWO "OBJECTS" ARE SIMILAR.

IF THE SIMILARITY REACH A FIXED t (TRESHOLD) THE OBJECTS ARE CONSIDERED THE SAME OBJECT...

- MODULES

THE BIOMETRIC SYSTEMS ARE DIVIDED IN 4 BLOCKS



- SENSOR MODULE

IT CAPTURES RAW DATA.

- FEATURE EXTRACTOR MODULE

ESTRACT VECTOR AND TEMPLATE FROM RAW DATA.

- MATCHING MODULE

IF THE RAW DATA IS A PROBE IT COMPARE DATA (TEMPLATE) WITH GALLERY.

- DECISIONAL MODULE

IT MAKES A DECISION.

- CHECK AND IDENTIFICATION